Experiment:21

```
mysql> SELECT CONCAT('Hello', SPACE(3), 'World') AS Result;
+---------------+
| Result        |
+---------------+
| Hello   World |
+---------------+
1 row in set (0.04 sec)

mysql> SELECT SUBSTR('COMPUTER', 1, 4) AS Result;
+--------+
| Result |
+--------+
| COMP   |
+--------+
1 row in set (0.00 sec)

mysql> SELECT UPPER('hello world') AS Result;
+-------------+
| Result      |
+-------------+
| HELLO WORLD |
+-------------+
1 row in set (0.00 sec)

mysql> SELECT LOWER('HELLO WORLD') AS Result;
+-------------+
| Result      |
+-------------+
| hello world |
+-------------+
1 row in set (0.04 sec)
```

```
mysql> SELECT TRIM('   MYSQL   ') AS Result;
+--------+
| Result |
+--------+
| MYSQL  |
+--------+
1 row in set (0.00 sec)

mysql> SELECT LENGTH('COMPUTER') AS Result;
+--------+
| Result |
+--------+
|      8 |
+--------+
1 row in set (0.00 sec)
```

Experiment:22

```
mysql> CREATE TABLE trigger_student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30),
    ->     marks INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO trigger_student VALUES
    -> (1, 'Ravi', 85),
    -> (2, 'Priya', 90);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE student_audit (
    ->     message VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER student_before_insert
    -> BEFORE INSERT ON trigger_student
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO student_audit
    ->     VALUES ('New student record is being inserted');
    -> END //
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO trigger_student VALUES
    -> (3, 'Arun', 75);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> SELECT * FROM trigger_student;
+----+-------+-------+
| id | name  | marks |
+----+-------+-------+
|  1 | Ravi  |    85 |
|  2 | Priya |    90 |
|  3 | Arun  |    75 |
+----+-------+-------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM student_audit;
+--------------------------------------+
| message                              |
+--------------------------------------+
| New student record is being inserted |
+--------------------------------------+
1 row in set (0.00 sec)

mysql> DROP TRIGGER student_before_insert;
Query OK, 0 rows affected (0.01 sec)
```

Experiment:23

```
mysql> USE college;
Database changed
mysql> SHOW TABLES;
+-------------------+
| Tables_in_college |
+-------------------+
| admission         |
| bus               |
| bus_ticket        |
| department        |
| reservation       |
| student           |
| student_auto      |
| student_log       |
| student_view      |
| train             |
+-------------------+
10 rows in set (0.00 sec)

mysql> DESC train;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| train_no    | int          | NO   | PRI | NULL    |       |
| train_name  | varchar(50)  | YES  |     | NULL    |       |
| source      | varchar(30)  | YES  |     | NULL    |       |
| destination | varchar(30)  | YES  |     | NULL    |       |
| seats       | int          | YES  |     | NULL    |       |
| fare        | decimal(8,2) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
6 rows in set (0.01 sec)

mysql> DESC reservation;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| ticket_id      | int          | NO   | PRI | NULL    | auto_increment |
| passenger_name | varchar(50)  | YES  |     | NULL    |                |
| train_no       | int          | YES  | MUL | NULL    |                |
| seats_booked   | int          | YES  |     | NULL    |                |
| total_fare     | decimal(8,2) | YES  |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM train;
+----------+----------------+---------+-------------+-------+--------+
| train_no | train_name     | source  | destination | seats | fare   |
+----------+----------------+---------+-------------+-------+--------+
|      101 | Chennai Express | Chennai | Bangalore   |    50 | 450.00 |
|      102 | Pandian Express | Chennai | Madurai     |    40 | 350.00 |
|      103 | Kovai Express   | Chennai | Coimbatore  |    30 | 400.00 |
+----------+----------------+---------+-------------+-------+--------+
3 rows in set (0.05 sec)

mysql> SELECT * FROM reservation;
+-----------+----------------+----------+-------------+------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare |
+-----------+----------------+----------+-------------+------------+
|         1 | Ravi           |      101 |            2 |     900.00 |
+-----------+----------------+----------+-------------+------------+
1 row in set (0.05 sec)

mysql> DESC train;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| train_no    | int          | NO   | PRI | NULL    |       |
| train_name  | varchar(50)  | YES  |     | NULL    |       |
| source      | varchar(30)  | YES  |     | NULL    |       |
| destination | varchar(30)  | YES  |     | NULL    |       |
| seats       | int          | YES  |     | NULL    |       |
| fare        | decimal(8,2) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> DESC reservation;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| ticket_id      | int          | NO   | PRI | NULL    | auto_increment |
| passenger_name | varchar(50)  | YES  |     | NULL    |                |
| train_no       | int          | YES  | MUL | NULL    |                |
| seats_booked   | int          | YES  |     | NULL    |                |
| total_fare     | decimal(8,2) | YES  |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE reservation
    -> ADD booking_source VARCHAR(30);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC reservation;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| ticket_id      | int          | NO   | PRI | NULL    | auto_increment |
| passenger_name | varchar(50)  | YES  |     | NULL    |                |
| train_no       | int          | YES  | MUL | NULL    |                |
| seats_booked   | int          | YES  |     | NULL    |                |
| total_fare     | decimal(8,2) | YES  |     | NULL    |                |
| booking_source | varchar(30)  | YES  |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)

mysql> INSERT INTO reservation
    -> (passenger_name, train_no, seats_booked, total_fare, booking_source)
    -> VALUES
    -> ('Meena', 102, 2, 700.00, 'WhatsApp');
Query OK, 1 row affected (0.04 sec)

mysql> INSERT INTO reservation
    -> (passenger_name, train_no, seats_booked, total_fare, booking_source)
    -> VALUES
    -> ('Divya', 103, 3, 1200.00, 'Instagram');
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM reservation;
+-----------+----------------+----------+-------------+------------+----------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare | booking_source |
+-----------+----------------+----------+-------------+------------+----------------+
|         1 | Ravi           |      101 |            2 |     900.00 | NULL           |
|         2 | Meena          |      102 |            2 |     700.00 | WhatsApp       |
|         3 | Divya          |      103 |            3 |    1200.00 | Instagram      |
+-----------+----------------+----------+-------------+------------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM reservation
    -> WHERE booking_source = 'WhatsApp';
+-----------+----------------+----------+-------------+------------+----------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare | booking_source |
+-----------+----------------+----------+-------------+------------+----------------+
|         2 | Meena          |      102 |            2 |     700.00 | WhatsApp       |
+-----------+----------------+----------+-------------+------------+----------------+
```

```
mysql> SELECT
    -> reservation.ticket_id,
    -> reservation.passenger_name,
    -> train.train_name,
    -> train.source,
    -> train.destination,
    -> reservation.seats_booked,
    -> reservation.total_fare,
    -> reservation.booking_source
    -> FROM reservation
    -> JOIN train
    -> ON reservation.train_no = train.train_no;
+-----------+----------------+-----------------+---------+-------------+--------------+------------+--
--------------+
| ticket_id | passenger_name | train_name      | source  | destination | seats_booked | total_fare | b
ooking_source |
+-----------+----------------+-----------------+---------+-------------+--------------+------------+--
--------------+
|         1 | Ravi           | Chennai Express | Chennai | Bangalore   |            2 |     900.00 | N
ULL           |
|         2 | Meena          | Pandian Express | Chennai | Madurai     |            2 |     700.00 | W
hatsApp       |
|         3 | Divya          | Kovai Express   | Chennai | Coimbatore  |            3 |    1200.00 | I
nstagram      |
+-----------+----------------+-----------------+---------+-------------+--------------+------------+--
--------------+
3 rows in set (0.00 sec)

mysql> SELECT
    -> passenger_name,
    -> seats_booked,
    -> total_fare,
    -> (total_fare / seats_booked) AS fare_per_seat
    -> FROM reservation;
+----------------+--------------+------------+---------------+
| passenger_name | seats_booked | total_fare | fare_per_seat |
+----------------+--------------+------------+---------------+
| Ravi           |            2 |     900.00 |    450.000000 |
| Meena          |            2 |     700.00 |    350.000000 |
| Divya          |            3 |    1200.00 |    400.000000 |
+----------------+--------------+------------+---------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM reservation
    -> WHERE booking_source = 'Instagram';
+-----------+----------------+----------+--------------+------------+----------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare | booking_source |
+-----------+----------------+----------+--------------+------------+----------------+
|         3 | Divya          |      103 |            3 |    1200.00 | Instagram      |
+-----------+----------------+----------+--------------+------------+----------------+
```

```
mysql> SELECT
    -> booking_source,
    -> COUNT(*) AS total_bookings
    -> FROM reservation
    -> GROUP BY booking_source;
+----------------+----------------+
| booking_source | total_bookings |
+----------------+----------------+
| NULL           |              1 |
| WhatsApp       |              1 |
| Instagram      |              1 |
+----------------+----------------+
3 rows in set (0.04 sec)

mysql> SELECT * FROM reservation
    -> ORDER BY total_fare DESC;
+-----------+----------------+----------+--------------+------------+----------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare | booking_source |
+-----------+----------------+----------+--------------+------------+----------------+
|         3 | Divya          |      103 |            3 |    1200.00 | Instagram      |
|         1 | Ravi           |      101 |            2 |     900.00 | NULL           |
|         2 | Meena          |      102 |            2 |     700.00 | WhatsApp       |
+-----------+----------------+----------+--------------+------------+----------------+
3 rows in set (0.00 sec)

mysql> UPDATE reservation
    -> SET booking_source = 'Facebook'
    -> WHERE ticket_id = 1;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM reservation;
+-----------+----------------+----------+--------------+------------+----------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare | booking_source |
+-----------+----------------+----------+--------------+------------+----------------+
|         1 | Ravi           |      101 |            2 |     900.00 | Facebook       |
|         2 | Meena          |      102 |            2 |     700.00 | WhatsApp       |
|         3 | Divya          |      103 |            3 |    1200.00 | Instagram      |
+-----------+----------------+----------+--------------+------------+----------------+
3 rows in set (0.00 sec)

mysql> DELETE FROM reservation
    -> WHERE ticket_id = 3;
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM reservation;
+-----------+----------------+----------+--------------+------------+----------------+
| ticket_id | passenger_name | train_no | seats_booked | total_fare | booking_source |
+-----------+----------------+----------+--------------+------------+----------------+
|         1 | Ravi           |      101 |            2 |     900.00 | Facebook       |
|         2 | Meena          |      102 |            2 |     700.00 | WhatsApp       |
+-----------+----------------+----------+--------------+------------+----------------+
2 rows in set (0.00 sec)
```

Experiment:24

```
mysql> DESC admission;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| application_id   | int         | NO   | PRI | NULL    |       |
| student_name     | varchar(50) | YES  |     | NULL    |       |
| course           | varchar(50) | YES  |     | NULL    |       |
| marks            | int         | YES  |     | NULL    |       |
| phone            | varchar(15) | YES  |     | NULL    |       |
| admission_status | varchar(20) | YES  |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM admission;
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              1 | Ravi         | CSE    |    95 | 9876543210 | Approved         |
|              2 | Priya        | IT     |    90 | 9876543211 | Approved         |
|              3 | Arun         | ECE    |    65 | 9876543212 | Approved         |
+----------------+--------------+--------+-------+------------+------------------+
3 rows in set (0.04 sec)

mysql> INSERT INTO admission
    -> (application_id, student_name, course, marks, phone, admission_status)
    -> VALUES
    -> (4, 'Meghana', 'CSE', 88, '9000000001', 'Pending'),
    -> (5, 'Kiran', 'IT', 82, '9000000002', 'Pending');
Query OK, 2 rows affected (0.04 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM admission;
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              1 | Ravi         | CSE    |    95 | 9876543210 | Approved         |
|              2 | Priya        | IT     |    90 | 9876543211 | Approved         |
|              3 | Arun         | ECE    |    65 | 9876543212 | Approved         |
|              4 | Meghana      | CSE    |    88 | 9000000001 | Pending          |
|              5 | Kiran        | IT     |    82 | 9000000002 | Pending          |
+----------------+--------------+--------+-------+------------+------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM admission
    -> WHERE student_name = 'Priya';
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              2 | Priya        | IT     |    90 | 9876543211 | Approved         |
+----------------+--------------+--------+-------+------------+------------------+
```

```
mysql> SELECT * FROM admission
    -> WHERE course = 'CSE';
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              1 | Ravi         | CSE    |    95 | 9876543210 | Approved         |
|              4 | Meghana      | CSE    |    88 | 9000000001 | Pending          |
+----------------+--------------+--------+-------+------------+------------------+
2 rows in set (0.00 sec)

mysql> UPDATE admission
    -> SET admission_status = 'Approved'
    -> WHERE application_id = 4;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM admission
    -> WHERE application_id = 4;
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              4 | Meghana      | CSE    |    88 | 9000000001 | Approved         |
+----------------+--------------+--------+-------+------------+------------------+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS total_applications
    -> FROM admission;
+--------------------+
| total_applications |
+--------------------+
|                  5 |
+--------------------+
1 row in set (0.05 sec)

mysql> SELECT * FROM admission
    -> WHERE marks > 80;
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              1 | Ravi         | CSE    |    95 | 9876543210 | Approved         |
|              2 | Priya        | IT     |    90 | 9876543211 | Approved         |
|              4 | Meghana      | CSE    |    88 | 9000000001 | Approved         |
|              5 | Kiran        | IT     |    82 | 9000000002 | Pending          |
```

```
mysql>
mysql> SELECT * FROM admission
    -> ORDER BY marks DESC;
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              1 | Ravi         | CSE    |    95 | 9876543210 | Approved         |
|              2 | Priya        | IT     |    90 | 9876543211 | Approved         |
|              4 | Meghana      | CSE    |    88 | 9000000001 | Approved         |
|              5 | Kiran        | IT     |    82 | 9000000002 | Pending          |
|              3 | Arun         | ECE    |    65 | 9876543212 | Approved         |
+----------------+--------------+--------+-------+------------+------------------+
5 rows in set (0.00 sec)

mysql> SELECT course, COUNT(*) AS total_students
    -> FROM admission
    -> GROUP BY course;
+--------+----------------+
| course | total_students |
+--------+----------------+
| CSE    |              2 |
| IT     |              2 |
| ECE    |              1 |
+--------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM admission
    -> WHERE admission_status = 'Pending';
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              5 | Kiran        | IT     |    82 | 9000000002 | Pending          |
+----------------+--------------+--------+-------+------------+------------------+
1 row in set (0.00 sec)

mysql> DELETE FROM admission
    -> WHERE application_id = 5;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM admission;
+----------------+--------------+--------+-------+------------+------------------+
| application_id | student_name | course | marks | phone      | admission_status |
+----------------+--------------+--------+-------+------------+------------------+
|              1 | Ravi         | CSE    |    95 | 9876543210 | Approved         |
|              2 | Priya        | IT     |    90 | 9876543211 | Approved         |
|              3 | Arun         | ECE    |    65 | 9876543212 | Approved         |
|              4 | Meghana      | CSE    |    88 | 9000000001 | Approved         |
+----------------+--------------+--------+-------+------------+------------------+
4 rows in set (0.00 sec)
```

Experiment:25

```
mysql> DESC bus;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| bus_id          | int          | NO   | PRI | NULL    |       |
| bus_name        | varchar(50)  | YES  |     | NULL    |       |
| source          | varchar(30)  | YES  |     | NULL    |       |
| destination     | varchar(30)  | YES  |     | NULL    |       |
| fare            | decimal(8,2) | YES  |     | NULL    |       |
| available_seats | int          | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
6 rows in set (0.01 sec)

mysql> DESC bus_ticket;
+----------------+--------------+------+-----+-------------------+-------------------+
| Field          | Type         | Null | Key | Default           | Extra             |
+----------------+--------------+------+-----+-------------------+-------------------+
| ticket_id      | int          | NO   | PRI | NULL              | auto_increment    |
| passenger_name | varchar(50)  | YES  |     | NULL              |                   |
| bus_id         | int          | YES  | MUL | NULL              |                   |
| seats_booked   | int          | YES  |     | NULL              |                   |
| total_fare     | decimal(8,2) | YES  |     | NULL              |                   |
| booking_time   | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| qr_code        | varchar(100) | YES  |     | NULL              |                   |
+----------------+--------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)

mysql> SELECT * FROM bus;
+--------+--------------+---------+-------------+-------+-----------------+
| bus_id | bus_name     | source  | destination | fare  | available_seats |
+--------+--------------+---------+-------------+-------+-----------------+
|    101 | City Express | Chennai | Tambaram    | 30.00 |              38 |
|    102 | Green Line   | Chennai | Velachery   | 25.00 |              35 |
|    103 | Metro Bus    | Chennai | Guindy      | 20.00 |              30 |
+--------+--------------+---------+-------------+-------+-----------------+
3 rows in set (0.05 sec)

mysql> SELECT * FROM bus_ticket;
+-----------+----------------+--------+--------------+------------+---------------------+-------------
--+
| ticket_id | passenger_name | bus_id | seats_booked | total_fare | booking_time        | qr_code
  |
+-----------+----------------+--------+--------------+------------+---------------------+-------------
--+
|         1 | Priya          |    101 |            2 |      60.00 | 2026-10-26 22:56:02 | QR-TICKET-00
1 |
+-----------+----------------+--------+--------------+------------+---------------------+-------------
--+
1 row in set (0.01 sec)
```

```
mysql> INSERT INTO bus_ticket
    -> (passenger_name, bus_id, seats_booked, total_fare, qr_code)
    -> VALUES
    -> ('Nandhini', 102, 2, 50.00, 'QR-BUS-002');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO bus_ticket
    -> (passenger_name, bus_id, seats_booked, total_fare, qr_code)
    -> VALUES
    -> ('Ramesh', 103, 3, 60.00, 'QR-BUS-003');
Query OK, 1 row affected (0.05 sec)

mysql> ^C
mysql> SELECT * FROM bus_ticket;
+-----------+----------------+--------+--------------+------------+---------------------+-------------
--+
| ticket_id | passenger_name | bus_id | seats_booked | total_fare | booking_time        | qr_code
  |
+-----------+----------------+--------+--------------+------------+---------------------+-------------
--+
|         1 | Priya          |    101 |            2 |      60.00 | 2026-10-26 22:56:02 | QR-TICKET-00
1 |
|         2 | Nandhini       |    102 |            2 |      50.00 | 2026-11-03 13:55:15 | QR-BUS-002
  |
|         3 | Ramesh         |    103 |            3 |      60.00 | 2026-11-03 13:55:39 | QR-BUS-003
  |
+-----------+----------------+--------+--------------+------------+---------------------+-------------
--+
3 rows in set (0.00 sec)

mysql> UPDATE bus
    -> SET available_seats = available_seats - 2
    -> WHERE bus_id = 102;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE bus
    -> SET available_seats = available_seats - 3
    -> WHERE bus_id = 103;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT bus_id, bus_name, available_seats
    -> FROM bus;
+--------+-------------+-----------------+
| bus_id | bus_name    | available_seats |
+--------+-------------+-----------------+
|    101 | City Express |              38 |
|    102 | Green Line   |              33 |
|    103 | Metro Bus    |              27 |
+--------+-------------+-----------------+
```

```
mysql> SELECT * FROM bus
    -> WHERE available_seats > 0;
+--------+--------------+---------+-------------+-------+-----------------+
| bus_id | bus_name     | source  | destination | fare  | available_seats |
+--------+--------------+---------+-------------+-------+-----------------+
|    101 | City Express | Chennai | Tambaram    | 30.00 |              38 |
|    102 | Green Line   | Chennai | Velachery   | 25.00 |              33 |
|    103 | Metro Bus    | Chennai | Guindy      | 20.00 |              27 |
+--------+--------------+---------+-------------+-------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT
    -> bus_name,
    -> fare,
    -> 3 AS seats,
    -> (fare * 3) AS total_fare
    -> FROM bus
    -> WHERE bus_id = 102;
+------------+-------+-------+------------+
| bus_name   | fare  | seats | total_fare |
+------------+-------+-------+------------+
| Green Line | 25.00 |     3 |      75.00 |
+------------+-------+-------+------------+
1 row in set (0.04 sec)

mysql> SELECT * FROM bus_ticket
    -> WHERE qr_code = 'QR-BUS-002';
+-----------+----------------+--------+-------------+------------+---------------------+------------+
| ticket_id | passenger_name | bus_id | seats_booked | total_fare | booking_time        | qr_code    |
+-----------+----------------+--------+-------------+------------+---------------------+------------+
|         2 | Nandhini       |    102 |            2 |      50.00 | 2026-11-03 13:55:15 | QR-BUS-002 |
+-----------+----------------+--------+-------------+------------+---------------------+------------+
1 row in set (0.00 sec)

mysql> SELECT
    -> bus_ticket.ticket_id,
    -> bus_ticket.passenger_name,
    -> bus.bus_name,
    -> bus.source,
    -> bus.destination,
    -> bus_ticket.seats_booked,
    -> bus_ticket.total_fare,
    -> bus_ticket.qr_code
    -> FROM bus_ticket
    -> JOIN bus
    -> ON bus_ticket.bus_id = bus.bus_id;
```

```
    -> ON bus_ticket.bus_id = bus.bus_id;
+-----------+----------------+--------------+---------+-------------+--------------+------------+---------------+
| ticket_id | passenger_name | bus_name     | source  | destination | seats_booked | total_fare | qr_code       |
+-----------+----------------+--------------+---------+-------------+--------------+------------+---------------+
|         1 | Priya          | City Express | Chennai | Tambaram    |            2 |      60.00 | QR-TICKET-001 |
|         2 | Nandhini       | Green Line   | Chennai | Velachery   |            2 |      50.00 | QR-BUS-002    |
|         3 | Ramesh         | Metro Bus    | Chennai | Guindy      |            3 |      60.00 | QR-BUS-003    |
+-----------+----------------+--------------+---------+-------------+--------------+------------+---------------+
3 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS total_bookings
    -> FROM bus_ticket;
+----------------+
| total_bookings |
+----------------+
|              3 |
+----------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM bus_ticket
    -> ORDER BY total_fare DESC;
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
| ticket_id | passenger_name | bus_id | seats_booked | total_fare | booking_time        | qr_code       |
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
|         1 | Priya          |    101 |            2 |      60.00 | 2026-10-26 22:56:02 | QR-TICKET-001 |
|         3 | Ramesh         |    103 |            3 |      60.00 | 2026-11-03 13:55:39 | QR-BUS-003    |
|         2 | Nandhini       |    102 |            2 |      50.00 | 2026-11-03 13:55:15 | QR-BUS-002    |
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
3 rows in set (0.00 sec)

mysql> UPDATE bus
    -> SET available_seats = available_seats + 3
    -> WHERE bus_id = 103;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM bus_ticket;
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
| ticket_id | passenger_name | bus_id | seats_booked | total_fare | booking_time        | qr_code       |
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
|         1 | Priya          |    101 |            2 |      60.00 | 2026-10-26 22:56:02 | QR-TICKET-001 |
|         2 | Nandhini       |    102 |            2 |      50.00 | 2026-11-03 13:55:15 | QR-BUS-002    |
|         3 | Ramesh         |    103 |            3 |      60.00 | 2026-11-03 13:55:39 | QR-BUS-003    |
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
3 rows in set (0.00 sec)

mysql> DELETE FROM bus_ticket
    -> WHERE ticket_id = 3;
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM bus_ticket;
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
| ticket_id | passenger_name | bus_id | seats_booked | total_fare | booking_time        | qr_code       |
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
|         1 | Priya          |    101 |            2 |      60.00 | 2026-10-26 22:56:02 | QR-TICKET-001 |
|         2 | Nandhini       |    102 |            2 |      50.00 | 2026-11-03 13:55:15 | QR-BUS-002    |
+-----------+----------------+--------+--------------+------------+---------------------+---------------+
2 rows in set (0.00 sec)
```